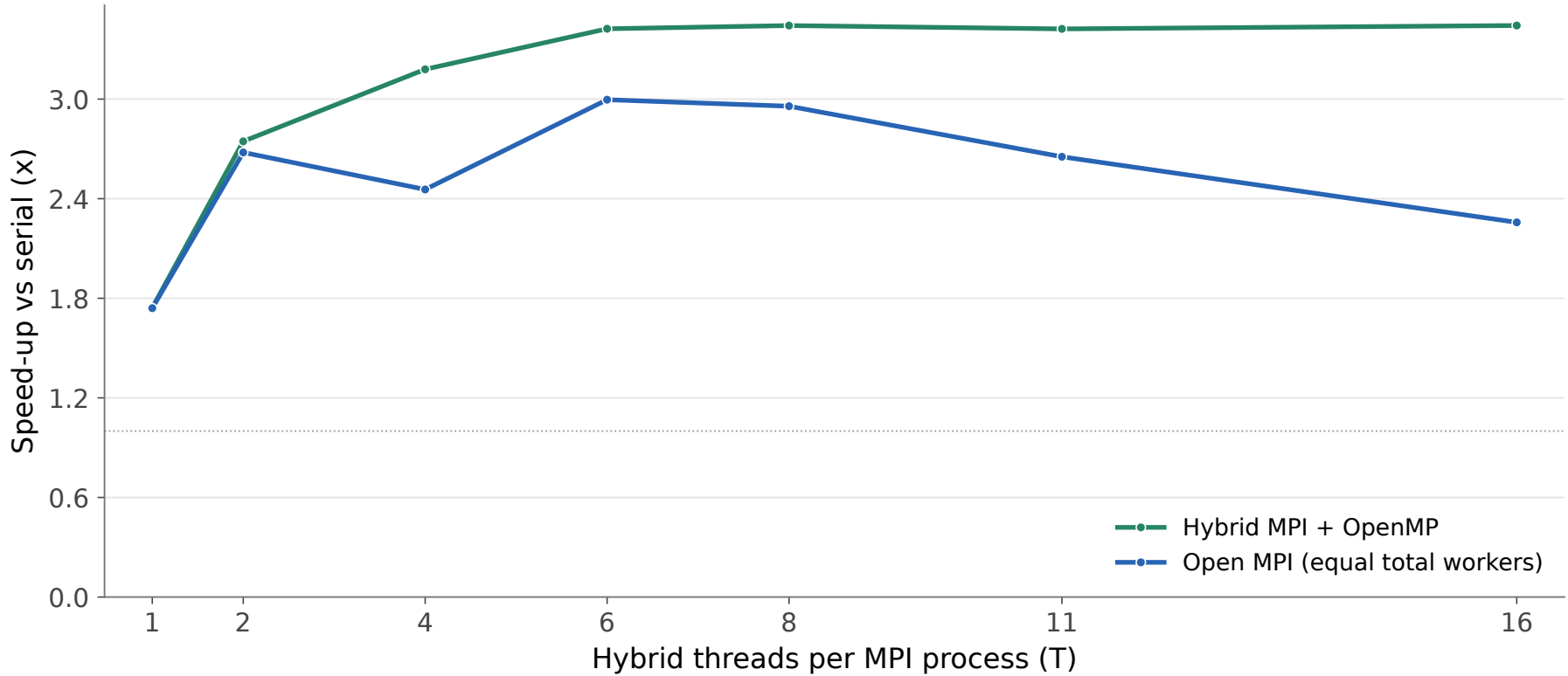


4. Speed-up: hybrid vs MPI

Hybrid: 2 MPI processes x T threads | MPI-only: 2 x T processes



Each point: median speed-up across 32 n values, 2.5-10 million; median of 3 runs at each n.

Overall wall-clock time: setup + search + gather (MPI) + file output; MPI startup excluded.

Apple M3 Pro | 11 CPU cores | Local measurements